



Department of Information Technology
Academic Year 2025 – 2026 (Even Semester)

Year, Degree, & Branch : VI-Semester-, B.Tech-IT
Course Code & Title : CCS374- & Web Application Security
Name of the Faculty member : B.Thevahi, AP/IT
Innovative Practice Description

- **Unit / Topic** : III/ Token Based Authentication
- **Course Outcome** : CO3
- **Activity Chosen** : Think_Pair_Share
- **Justification:**

Think–Pair–Share helps learners actively engage with the topic by first thinking individually, then discussing ideas with a partner, and finally sharing insights with the class. Here, we used the Think–Pair–Share activity for Token-Based Authentication because it helps students understand the token mechanism and the process involved in token generation, validation, and usage clearly.

- **Time Allotted for the Activity:** 15 Minutes

❖ **Details of the Implementation:**

- The Instructor took a class on Token-Based Authentication and explained its concepts, working process, components, advantages, and differences from session-based authentication
- Then the Instructor instructed the students to think individually about how token-based authentication works and told to write key points.
- The students then paired with neighboring students and discussed their understanding, compared their answers that was shown in Figure1.
- After the discussion, each pair shared their ideas with the class as shown in the Figure2.

• **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO5	PO10	PO12	PSO1	PSO3
CO3	1	2	3	2	2	2	2	2

(1 – Low

2 – Moderate

3 – High)

PO / PSO mapped:

Innovative practice	PO1	PO2	PO3	PO5	PO10	PO12	PSO1	PSO3
CO1	1	2	3	2	2	2	2	2
Justification for correlation	Students understand and apply the basic concepts of token-based authentication, including access and refresh tokens..	Students analyze authentication-related security threats such as token theft and unauthorized access.	The students explore token generation and validation processes.	Students gain knowledge of API authentication and bearer tokens.	The activity improves students' ability to communicate technical concepts related to authentication and cybersecurity.	Students are motivated to update themselves with evolving authentication standards.	Students develop domain knowledge in implementing secure authentication mechanisms.	Students understand how token-based authentication supports secure and scalable software development..

• **Images / Screenshot of the practice:**



Figure1: Think_Pair_Share Activity



Figure2: Presentation by A.Poornima and M.Valli

- **Reflective Critique:**

- ❖ ***Feedback of practice from students and other stakeholders:***

- All students eagerly participated in thinking about and discussing token-based authentication and its verification process.
- During pair discussions, students shared real-time examples of authentication systems used in web and mobile applications, which improved their practical understanding
- Many students said that explaining concepts like access tokens and refresh tokens to their partners increased their confidence and cleared their doubts.

- ❖ ***Benefit of the practice:***

- The Think–Pair–Share activity helps students understand token-based authentication concepts clearly through individual thinking and peer discussion.
- It improves students' critical thinking, communication skills, and confidence in explaining modern authentication mechanisms.
- The practice increases student involvement and interest during the learning process.

- ❖ ***Challenges faced in implementation:***

- Some students initially found it difficult to understand technical terms related to token-based authentication, such as access tokens and refresh tokens
- A few students were hesitant to participate actively in discussions during the first implementation of the Think–Pair–Share activity.
- Time management was slightly challenging, as discussions required additional time for clarification of authentication concepts.

References:

1. <https://www.ritrjpm.ac.in/images/B.Tech-IT/2025-2026/2.%20Think%20Pair%20Share.pdf>
2. https://www.ritrjpm.ac.in/images/B.Tech-IT/20252026/5.GM_CD3291_Think%20Pair%20Share.pdf

Signature of Faculty Member

HOD